Use equivalent fractions when dividing

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________________ Date: _______________

1. Read the model: Calculate 5/6 divided by 3. Copy or complete the first step.

2. Calculate 2 1/4 divided by 3.

3. Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?

4. Miro says: Miro divides both the numerator and denominator by the whole number. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 5/6 divided by 3. Copy or complete the first step.
   Answer 5/18.

2. Calculate 2 1/4 divided by 3.
   Answer 3/4.

3. Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?
   Answer 7/24 litre.

4. Miro says: Miro divides both the numerator and denominator by the whole number. Is this correct? Explain.
   Answer Division changes the size of each share. Use a bar model or multiply by the unit fraction, then simplify.

Use equivalent fractions when dividing

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. Calculate 5/6 divided by 3.

2. Calculate 2 1/4 divided by 3.

3. Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?

4. Solve this independently and show every stage: Calculate 5/6 divided by 3.

5. Use a second method or representation for: Calculate 2 1/4 divided by 3.

6. Check this result using an inverse, estimate or known fact: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?

7. Prove or explain your reasoning: Calculate 1/3 divided by 4.

8. Identify and correct this misconception: Miro divides both the numerator and denominator by the whole number.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?

Teacher answers and checking prompts

1. Calculate 5/6 divided by 3.
   Answer 5/18.

2. Calculate 2 1/4 divided by 3.
   Answer 3/4.

3. Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?
   Answer 7/24 litre.

4. Solve this independently and show every stage: Calculate 5/6 divided by 3.
   Answer 5/18.

5. Use a second method or representation for: Calculate 2 1/4 divided by 3.
   Answer Expected result: 3/4. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?
   Answer Expected result: 7/24 litre. A valid check must be shown.

7. Prove or explain your reasoning: Calculate 1/3 divided by 4.
   Answer 1/12. The explanation must show why the method works.

8. Identify and correct this misconception: Miro divides both the numerator and denominator by the whole number.
   Answer Division changes the size of each share. Use a bar model or multiply by the unit fraction, then simplify.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use equivalent fractions when dividing

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Calculate 1/3 divided by 4.

2. Prove the answer to this question, rather than only calculating it: Calculate 5/6 divided by 3.

3. Solve this in two different ways and compare them: Calculate 2 1/4 divided by 3.

4. Work backwards from the answer to reconstruct the question: 7/24 litre.

5. Explain why this reasoning fails: Miro divides both the numerator and denominator by the whole number.

6. Create a new example that tests the same idea as: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate 1/3 divided by 4.
   Answer 1/12. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Calculate 5/6 divided by 3.
   Answer Expected result: 5/18. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 2 1/4 divided by 3.
   Answer Expected result: 3/4. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 7/24 litre.
   Answer One valid reconstruction is: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro divides both the numerator and denominator by the whole number.
   Answer Division changes the size of each share. Use a bar model or multiply by the unit fraction, then simplify.

6. Create a new example that tests the same idea as: Seven eighths of a litre is shared equally between 3 cups. How much is in each cup?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.